

Lesson 29 — Change the Pose Program

Matches LEGO Teacher Portal lesson: "Rainforest Rhythm".

Objective (I can...): I can change a pose program to do what I want, and tell one reason the AI might guess wrong. **CS concept:** Modifying programs that use pretrained AI pose blocks; example quality and fairness · **Builds on:** Lesson 28 (using pose blocks). *Honest framing: kids modify a program that uses ready-made pose blocks and watch the coach's demo learn from examples — they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved.*

AI Connection (2 min, read aloud): This is why we said "more tries help" back in loops — a computer learns poses from lots of different examples, and it can still guess wrong. *Ask the group:* "What kind of example would help a computer that keeps mixing up two poses?"

Materials: LEGO CS & AI kit + device per pair with camera, a big screen for the coach's Teachable Machine demo. **Model reuse:** AI & Data model from Lesson 26 — code only.

60-minute flow - Warm-up (10): "Tricky twins" — coach shows two poses that look alike; ask how you'd tell them apart for sure. Connect to clearer, more varied examples. - **Concept + demo (10):** On the **coach's** screen, run Google Teachable Machine: kids take turns showing example poses to the camera, the coach captures them, and the group watches the demo learn to tell poses apart — then sees it guess wrong when a pose only had one kind of example. **Say it plainly: the coach's computer is learning from the examples WE show it; the kids are exploring how it learns, not training their own AI.** On-device, no logins, pictures not saved. - **Build/Code (22):** Back at their own model, pairs **modify** their pose program from L28 — add or change a pose reaction, swap which pose does what — and test where it still guesses wrong. They keep a tally of "how many examples per pose" on the plan mat and ask "are our poses even?" *Wiggle break + swap at ~12 min.* - **Challenge (7):** Try a pose the block mixes up; talk about why it's tricky. - **Share + discussion (10):** Pairs report what they changed, then the group discusses **why** a computer still guesses wrong sometimes: maybe it only saw one kind of example, or one pose had way more examples than the other. Name one better example that would help, and talk about being fair — did the examples include all kinds of kids and poses? Model stays built.

Pair roles: Driver modifies the program, Navigator poses and reports right/wrong. Swap at the wiggle break. (*Trio: third keeps the example tally.*)

Differentiation: *Support* — change just one pose reaction. *Extension / fast-finisher* — add a third pose and balance the examples in the coach demo. *Reteach* — add one example at a time on the coach demo so the improvement is visible each time.

Exit check: Pair modifies a pose program so it does what they want, and can name one reason a computer might guess wrong and one way better examples would help. **At-home (optional):** Explain to a family member why a computer needs lots of different examples to learn poses well. ~5 min.